

1. Core Python skills that matter in production

Error handling & logging: Learn try/except patterns and the logging module. In production, failures should fail loudly and clearly, not silently. Working with structured data: JSON is essential (for AWS APIs, config files, and automation). YAML is worth learning too since tools like CloudFormation and Ansible rely on it. Environment management: Know how to use venv and pip properly. In production you'll often isolate dependencies. Requests & APIs: You're right to look at requests. Practically every AWS integration under the hood is API-based. Comfort with GET/POST, headers, authentication, and pagination will carry over directly.

2. AWS-specific Python skills

boto3 (AWS SDK for Python) is the big one. Start with simple tasks: list S3 buckets, upload a file, spin up/terminate an EC2 instance. Then move toward IAM, CloudWatch, etc. Being comfortable with boto3 makes you valuable immediately. awscli vs boto3: Sometimes you'll glue together AWS CLI commands with Python (subprocess), other times you'll use boto3 directly. Knowing when to use each is part of real-world judgment.

3. Broader "production-ready" practices

Idempotency: Make scripts safe to run multiple times without causing problems. (E.g., "create bucket if not exists" instead of blindly creating). Parameterization: Don't hardcode. Use config files, environment variables, or arguments so your script works across dev/test/prod. Testing: Even simple unit tests (pytest) help avoid nasty surprises. Packaging & reusability: Learn how to structure your code into functions and modules so it's maintainable.

4. What scripts are useful in the cloud world?

Think of tasks a sysadmin/cloud engineer does manually, and imagine automating them:

Rotate/expire IAM keys. Collect CloudWatch metrics or logs and push to Slack/Teams. Bulk tag or clean up unused EC2 volumes. Snapshot and prune old backups in S3. Run a daily health check across accounts and regions.

Each of those can be tackled with Python + boto3 once you're comfortable.

5. Transferable value

The truth is, you don't need to be a Python guru. What hiring managers want is someone who can:

Understand the AWS environment. Automate repetitive work. Write scripts that are safe, readable, and reliable.

You're already building the foundation. Focus on problem-driven learning: pick a small AWS task you'd normally do in the console and re-implement it in Python. Over time, those small scripts will give you both confidence and a portfolio you can show off.

I'll go first with boto3.

- Make script that will stop and start instance given a tag [Start: Yes]
- Then make the tag have the value [Start: 08:00]
- Then use boto3 to deploy the lambda and EventBridge to run it every hour.
- Then the script to nuke everything you just created.
- Then try to do it using cloudformation, then again using terraform. Finally using CDK.